Annex A: Calculation of circular angle spread

For a signal with *N* multi-paths, each has *M* sub-paths, the conventional angle spread calculation for the composite signal is given by

$$\sigma_{AS} = \sqrt{\frac{\sum_{n=1}^{N}\sum_{m=1}^{M}\left(\theta_{n,m,\mu}\right)^2 \cdot P_{n,m}}{\sum_{n=1}^{N}\sum_{m=1}^{M} P_{n,m}}} \qquad \text{(A-1)}$$

where $P_{n,m}$ is the power for the *m*th subpath of the *n*th path, $\theta_{n,m,\mu}$ is defined as

$$\theta_{n,m,\mu} = \mathrm{mod}(\theta_{n,m} - \mu_\theta + \pi,\ 2\pi) - \pi\,, \qquad \text{(A-2)}$$

μ_θ is defined as

$$\mu_\theta = \frac{\sum_{n=1}^{N}\sum_{m=1}^{M}\theta_{n,m} \cdot P_{n,m}}{\sum_{n=1}^{N}\sum_{m=1}^{M} P_{n,m}} \qquad \text{(A-3)}$$

and $\theta_{n,m}$ is the AoA (or AoD) of the *m*th subpath of the *n*th path. Note that we have dropped the AoA (AoD) subscript for convenience.

We note that the angle spread should be independent of a linear shift in the AoAs. In other words, by replacing $\theta_{n,m}$ with $\theta_{n,m} + \Delta$, the angle spread $\sigma_{AS}(\Delta)$ which is now a function of Δ should actually be constant no matter what Δ is. However, due to the ambiguity of the modulo 2π operation, this may not be the case. Therefore the angle spread should be the minimum of $\sigma_{AS}(\Delta)$ over all Δ:

$$\sigma_{AS} = \min_{\Delta}\ \sigma_{AS}(\Delta) = \sqrt{\frac{\sum_{n=1}^{N}\sum_{m=1}^{M}\left(\theta_{n,m,\mu}(\Delta)\right)^2 \cdot P_{n,m}}{\sum_{n=1}^{N}\sum_{m=1}^{M} P_{n,m}}} \qquad \text{(A-4)}$$

where $\theta_{n,m,\mu}(\Delta)$ is defined as

$$\theta_{n,m,\mu}(\Delta) = \mathrm{mod}(\theta_{n,m}(\Delta) - \mu_\theta(\Delta) + \pi,\ 2\pi) - \pi\,, \qquad \text{(A-5)}$$

$\mu_\theta(\Delta)$ is defined as

$$\mu_\theta(\Delta) = \frac{\sum_{n=1}^{N}\sum_{m=1}^{M}\theta_{n,m}(\Delta) \cdot P_{n,m}}{\sum_{n=1}^{N}\sum_{m=1}^{M} P_{n,m}} \qquad \text{(A-6)}$$

and $\theta_{n,m}(\Delta) = \mathrm{mod}(\theta_{n,m} + \Delta + \pi,\ 2\pi) - \pi$.